
Michael J. Lewis

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Quantitative researcher and e-trading strategist with over a decade of experience developing electronic trading, market-making, portfolio optimization, and quantitative investment solutions. Strong background in mathematics, statistical modeling, and software development with expertise across fixed income, multi-asset investing, and systematic research.

Work Experience

Bank of America Merrill Lynch

New York City, NY.

Vice President, E-Trading Quant

2023 - 2025

- Maintained & enhanced AMRS IG & HY desks' automated market making (AMM) algo coded in Python & Java
- Researched, developed, and implemented
 - factor-based models supporting pricing and market-making decisions, partnered with HY traders
 - parameter sensitivity tool for AMM
 - "order book" tool as risk control for AMM to address desk concerns of market crossing

Goldman Sachs Asset Management

New York City, NY.

Vice President, Multi Asset Solutions (MAS) Strat

2019 - 2023

- Lead integration of ESG data for benchmark / position analytics, and projects on ESG related research
- Project lead in Managed Accounts JV w/ Next Capital, as well as in-house glidepath analytics
- Researched quantitative trading signals
- Sped up in-house portfolio optimization tool from about a minute to seconds

Guggenheim Partners / Delaware Life

New York City, NY.

Vice President, Quantitative Researcher

2016 - 2018

- Designed and implemented high-performance systems in Perl and Python
- Consulted on risk management and performance metrics
- Built, mentored, and managed a team of quant researchers / strategists supporting investment research initiatives
- Researched literature, Machine Learning in particular, and developed investment strategies
- Co-authored with M. López de Prado:
"Detection of False Investment Strategies Using Unsupervised Learning Methods"

Courant Institute of Mathematical Sciences

New York City, NY.

Ph.D Candidate / Research Assistant under Jonathan Goodman

2010 - 2016

- Applied Markov Chain Monte Carlo methods, Bayesian analysis to the study of cryo-electron microscopy

Banc of America Securities

New York City, NY.

Financial Analyst / Associate in the RMBS Modeling Group

2005 - 2008

- Supported and enhanced mortgage prepay model for trading desk
 - Supported Modeling and Strategist groups for RMBS
 - Ran mortgage delinquency and prepay analysis for ad hoc trading desk requests
 - Led QA testing projects on DB
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Education

Courant Institute of Mathematical Sciences, NYU

New York City, NY

Ph.D. Mathematics

2016 (GPA: 3.9/4.0)

M.S. Mathematics

2012

Accolades: Henry M. MacCracken Fellowship, Member of American Mathematical Society (AMS)

Massachusetts Institute of Technology

Cambridge, MA

S.B. Mathematics, Minor in Economics

2004 (GPA: 4.4/5.0)

Related Coursework:

Numerical Methods / Analysis / Optimization, Monte Carlo Methods, PDEs, Numerical Methods for PDEs, Probability, Statistics, Mathematical Modeling, Stochastic Optimal Control, Financial Theory, Game Theory, International Trade, Behavioral Economics, Data Structures, Algorithms, High Performance Computing

Computer Skills / Other

- *Programming Languages:* Python (strong), Slang, Java, PERL, C++ (moderate)
- *Software:* iPython, Matlab, R, L^AT_EX, SQL / kdb
- *Skills:* Data Science, Math Modeling, High Performance Computing
- *Certifications:* Passed the Series 63 and Series 7 Exams (2006)